

ToddySoft Connect UMAS Driver - Ignition 8.1

Ignition 8.1 module for connecting to Schneider Electric **Modicon** controllers — M340, M580, Premium and Quantum — over UMAS, the native protocol Control Expert itself uses. Tags are addressed by the **symbol names from the PLC program**, not by register numbers: the driver downloads the controller's data dictionary and browses it.

UMAS and Modbus share port 502. They are multiplexed onto the same connection through different function codes, so a controller you can already reach with the Modbus module needs no extra port opening for this one. What they give you differs completely: Modbus offers numbered registers, UMAS offers the program's own symbols and types.

Installation

This module is for Ignition 8.1 only. It is built against the Ignition 8.1 SDK and will not run on any other major version. For Ignition 8.3, install the separate `...-8.3.mod1` build instead — the two are not interchangeable, and each gateway needs the build that matches it.

Ignition 8.0 and earlier are not supported and cannot be. The module is built on the device-driver API that Ignition introduced with 8.1, so on 8.0 the gateway fails to load it with `NoClassDefFoundError: ...opcua.server.api.AbstractDeviceModuleHook`.

Both builds carry the same name in the gateway's module list, so tell them apart by the file you installed or by the module description, which names the Ignition version.

The ToddySoft Connect modules are signed with a self-signed certificate (issued to *ToddySoft GmbH*) rather than a publicly accredited code-signing certificate. Ignition therefore asks you to review and accept that certificate once during installation - it is then remembered for all future ToddySoft modules signed with the same certificate.

1. Open the Ignition Gateway web interface and navigate to **Config > System > Modules**. Scroll to the bottom and click **Install or Upgrade a Module...**

2. Choose the `ToddySoft-Connect-UMAS-Driver-8.1.modl` file and click **Install**.
3. Ignition shows the self-signed certificate details (subject and issuer *ToddySoft GmbH*) together with a warning that the module was not written by Inductive Automation. Review the certificate, tick **I understand the risk and want to install this module** and click **Add Certificate and Install Module**.
4. The module is now trusted and listed under the **ToddySoft GmbH (self-signed certificate)** group with the state **Running**.

ToddySoft GmbH (self-signed certificate) View Certificate					
Name	Version	Description	License	State	
ToddySoft Connect CtrIX Driver	1.3.0-SNAPSHOT (b0)	ToddySoft Ignition driver for Bosch Rexroth CtrIX protocol. Built for Ignition 8.1.	Trial	Running	More ▾ restart
ToddySoft Connect S7 Driver	1.3.0-SNAPSHOT (b0)	ToddySoft Ignition driver for Siemens S7 protocol. Built for Ignition 8.1.	Trial	Running	More ▾ restart
ToddySoft Connect S7 Web API Driver	1.3.0-SNAPSHOT (b0)	ToddySoft Ignition driver for the Siemens SIMATIC Web API. A management-plane path for diagnostics, alarms and operating mode; use the S7 module for process data. Built for Ignition 8.1.	Trial	Running	More ▾ restart
ToddySoft Connect UMAS Driver	1.3.0-SNAPSHOT (b0)	ToddySoft Ignition driver for Schneider Electric UMAS protocol. Built for Ignition 8.1.	Trial	Running	More ▾ restart

Before you start: the controller side

- **The PLC program must be running.** The data dictionary is downloaded from the controller, so a stopped or empty program leaves nothing to browse and no symbol names to address.
- **UMAS must not be blocked.** Schneider controllers permit it by default, but some site security policies disable it — in which case the TCP connection succeeds and the protocol exchange does not.
- **Port 502 reachable** from the gateway host.

There are no credentials to configure: UMAS as this driver speaks it is unauthenticated, which is why this module has no username or password fields at all.

Creating a Device Connection

1. Go to **Config > OPC UA > Device Connections**.

Name	Type	Description	Enabled	Status	
CtrIX-Bench	ToddySoft Connect CtrIX Driver	Bosch Rexroth ctrIX on the bench	true	Connected	delete edit
Sorting-Lane-S7	ToddySoft Connect S7 Driver	S7-1200 on the sorting lane	true	Connected	delete edit
Sorting-Lane-WebAPI	ToddySoft Connect S7 Web API Driver	S7-1200 on the sorting lane	true	Connected	delete edit
→ Create new Device...					

2. Click **Create new Device...**, select **ToddySoft Connect UMAS Driver** as the device type and click **Next**.

ToddySoft Connect driver for Siemens S7 protocol

☐ **ToddySoft Connect S7 Web API Driver**
 ToddySoft Connect driver for the Siemens SIMATIC Web API (HTTPS) on S7-1200 and S7-1500. A management-plane path for diagnostics, alarms and operating mode - roughly an order of magnitude slower per tag than the S7 driver, with no change notification on the wire. Use the S7 driver for process data.

☒ **ToddySoft Connect UMAS Driver**
 ToddySoft Connect driver for Schneider Electric UMAS protocol

3. Configure the connection settings (see the sections below) and click **Save**.

4. The new device then appears in the Device Connections list.

Name	Type	Description	Enabled	Status	
CtrlX-Bench	ToddySoft Connect CtrlX Driver	Bosch Rexroth ctrlX on the bench	true	Connected	<button>delete</button> <button>edit</button>
Packaging-M580	ToddySoft Connect UMAS Driver	Modicon M580 on the packaging line	true	Connected	<button>delete</button> <button>edit</button>
Sorting-Lane-S7	ToddySoft Connect S7 Driver	S7-1200 on the sorting lane	true	Connected	<button>delete</button> <button>edit</button>
Sorting-Lane-WebAPI	ToddySoft Connect S7 Web API Driver	S7-1200 on the sorting lane	true	Connected	<button>delete</button> <button>edit</button>

→ [Create new Device...](#)

There is no device discovery for this driver, on either Ignition version, and no *Test Connection* action. Enter the host by hand, create the device, and read its status.

Setting blocks

The configuration is organized into categories: **General**, **Transport Settings**, **UMAS Settings** and **Audit Log Settings**.

General

Assigns a name (and optional description) to the connection and controls whether it is enabled.

General	
Name	Packaging-M580
Description	Modicon M580 on the packaging line
Enabled	☒ (default: true)

Setting	Description	Default
Name	Unique name for this device connection	
Description	Optional description for this connection	
Enabled	Whether the connection is active	true

This is the name the connection is then referenced under in other parts of Ignition.

Transport Settings

Transport Settings	
Host	192.168.24.31 Hostname or IP address of the Schneider Electric PLC
Port	502 TCP port number (default: 502) (default: 502)

Setting	Description	Default
Host	Hostname or IP address of the Schneider Electric PLC	
Port	TCP port number	502

There is no transport selector, and none is missing: UMAS runs over plain TCP, and there is no TLS variant to choose. Port 502 is the Modbus port, shared as described above.

UMAS Settings

UMAS Settings	
Request Timeout	5000 Default timeout for requests in milliseconds (default: 5.000)
Max Browse Depth	20 Maximum nesting depth the browser descends when expanding a symbol into child nodes. A type that refers back to one of its own ancestors forms a cycle the walk would otherwise follow until it runs out of memory. Real type hierarchies are a handful of levels deep (default: 20) (default: 20)
Browse Queries	** Query string for symbol browsing (default: ""). Separate multiple queries with ";" (default: "**")

Setting	Description	Default
Request Timeout	Milliseconds allowed for one request	5000
Max Browse Depth	How far the browser descends when expanding a symbol into children	20
Browse Queries	Query string for symbol browsing; separate multiple queries with ;	**

Max Browse Depth

This is a **cycle guard, not a performance knob**. A user-defined type that refers back to one of its own ancestors forms a loop the browse walk would otherwise follow until it runs out of memory. Real type hierarchies are a handful of levels deep, so the default of 20 sits far above anything legitimate and only ever stops a structure that could not terminate.

Browse Queries

The default `**` returns the whole data dictionary. On a controller with a large program that is a lot of nodes; narrow it to the symbols you actually want rather than browsing everything and filtering in the Designer.

Audit Log Settings

Audit Log Settings

Audit Log File

Path to the audit log file where events will be recorded. Leave empty to disable audit logging.

Setting	Description	Default
Audit Log File	Path to a file protocol audit events are written to; empty disables	

The audit log records protocol-level events and grows quickly, so enable it only while investigating a problem.

Tag Addressing

UMAS addresses are **symbolic** — the names from the PLC program — with dotted paths for struct fields and subscripts for array elements:

```
<symbol>(<field>)*([<index>(<index>)*])*
```

Address	Meaning
g_s16	A standalone <code>INT</code> variable
g_string1	A standalone <code>STRING</code>
MyStruct.RealField	A struct field
MyArray[5]	Element 5 of a one-dimensional array
MyMatrix[2][3]	Cell <code>[2][3]</code> of a two-dimensional array
MyStruct.Substruct.Field[10]	A nested struct path with array indexing

Names are case-sensitive and must match what Control Expert exported. Browse to discover them rather than typing them from the program listing — the browse returns the controller's actual data dictionary, which is the authority.

STRING buffer sizes are derived from the symbol table, not assumed, so a string reads at its real declared length.

After a Control Expert download

The browse result is cached and refreshed when the controller's program CRC changes. Immediately after downloading a changed program the cache can be one poll cycle stale; wait a cycle and browse again rather than concluding a symbol is missing.

Tag Groups

By default, all tags are collected in the default tag group, which means all selected tags are fetched at the default interval of once a second.

If different intervals are desired, create additional tag groups, configure the desired intervals there, and assign tags to the corresponding tag groups.

UMAS has no change notification on the wire, so subscriptions are served by polling. The tag group interval *is* the poll rate, and a fast group on many tags is real load on the controller.

Troubleshooting

Device shows "Faulted":

- Verify the PLC is reachable and that TCP port 502 is open through any site firewall
- Confirm the PLC program is running — without it there is no data dictionary
- Check whether a site security policy has disabled UMAS on the controller; the TCP connection then succeeds while the protocol exchange does not
- Review Gateway logs: **Status > Diagnostics > Logs**

INVALID_ADDRESS from a read or write:

- The symbol does not exist in the *current* program, or the dotted path / array index has a typo. Browse to confirm the actual name, and remember that names are case-sensitive

Nothing to browse, although the device connects:

- The program is not running, or the data dictionary has not finished downloading

A symbol is missing right after a Control Expert download:

- The cached browse is refreshed on a program-CRC poll. Wait one cycle

Tags show bad quality:

- Verify the symbol still exists in the program after the latest download
- Check spelling and case

Subscriptions miss fast changes:

- Expected: there is no change notification in UMAS, so subscriptions are polled.
Shorten the tag group interval, accepting the extra load

Docker related issues

The UMAS protocol carries no identity for this end of the connection, so a container's internal address is never announced to the controller and nothing has to be reconciled. A gateway in a bridged container connects normally, provided port 502 is routable.

VPN related issues

A remote controller reached over a VPN needs nothing special from this driver, for the same reason: the connection is an ordinary outbound TCP session. Only reachability matters.

Version history

Unreleased — 1.3.0

- First release of this manual.